



**College of Computing and Informatics
Department of Computer Science**

Course Syllabus

Course Title Junior Project in CS

**Course Code
1501394**

Spring 2025/2026

Course Syllabus

Course Title	Junior Project in CS				
Course Code	1501394				
Credit Hours	2	Contact Hours	2		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)	1501215 - Data Structures OR 1411215 - Data Structures OR 1501246 - Object Oriented Design With Java				
Co-requisite(s)					
Semester	Spring 2025/2026				
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Course Description:

The course serves as the first part of the one-year Senior Project in the BS Computer Science Program. Students work on a major computer science project integrating the knowledge gained from the courses in the curriculum. The project is team based. Students are expected to submit a proposal, followed by the detailed design of the project. Students are expected to create a working prototype of a software system and write a comprehensive project report. At the end, students present the current status of the project, demo the prototype and submit the final report. The main implementation of the project will continue in the Senior Project in Computer Science course.

Course Objectives/Goals:

The goals of this Course are to:

- Learn the guidelines for selecting the supervisor, selection of topic, and forming the team.
- Perform a comprehensive literature review, develop the ability to formulate new ideas different from the existing ideas, and develop the ability to think creatively.
- Improve the ability to plan and project a execute it, develop the ability to document a system, and present their work effectively both orally and in writing.
- Develop the ability to work in a team towards completing a major computer science project.

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Survey/research a project topic.
2. Collect, analyze, and select proper information to write a proposal for the project with system requirements.
3. Write progress and final report.
4. Present and demonstrate the design and prototype of the project.
5. Work as an efficient member of a team on the design and development of the project.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Survey/research a project topic. - Present and demonstrate the design and prototype of the project.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Present and demonstrate the design and prototype of the project. - Write progress and final report.
Function effectively as a member or leader of a team engaged in activities appropriate to the	- Work as an efficient member of a team on the design and

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
program's discipline.	development of the project.
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.	- Work as an efficient member of a team on the design and development of the project.
Communicate effectively in a variety of professional contexts.	- Present and demonstrate the design and prototype of the project. - Collect, analyze, and select proper information to write a proposal for the project with system requirements. - Write progress and final report. - Work as an efficient member of a team on the design and development of the project.
Apply computer science theory and software development fundamentals to produce computing-based solutions.	- Survey/research a project topic. - Present and demonstrate the design and prototype of the project. - Collect, analyze, and select proper information to write a proposal for the project with system requirements. - Work as an efficient member of a team on the design and development of the project.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	Introduction & Guidelines		1
2	Potential areas for the project(s)		1
3	How to give an oral presentation?		4
4	How to write a report?		3
5	Students work on a project	Project proposal	1,2,5
6	Students work on a project		1,2,5
7	Students work on a project		1,2,5
8	Students work on a project		1,2,3,5
9	Students work on a project	Progress report #1	1,2,5
10	Students work on a project		1,2,5
11	Students work on a project		1,2,5
12	Students work on a project		1,2,3,5
13	Students work on a project		1,2,3,5
14	Students work on a project		1,2,5
15	Students' presentations, critique, and demonstration.	Progress Report #2 + Presentation, final report and demo	3,4,5
16	Final Presentations		

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Examining Committee		25
Examiner	Week 15	25
Supervisor and Course Coordinator	Week 15	50
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Survey/research a project topic.	- Group work - Team based learning	- Examiner - Supervisor and Course Coordinator	70% of student achieve 70% of the CLO	- Critical and analytical thinking - Teamwork or collaboration	
Collect, analyze, and select proper information to write a proposal for the project with system requirements.	- Case studies - Group work	- Examiner - Examining Committee - Supervisor and Course Coordinator	70% of student achieve 70% of the CLO	- Intellectual curiosity and lifelong learning - Innovation / creative thinking - Autonomy and resiliency	
Write progress and final report.	- Problem-solving - Team based learning	- Examiner - Supervisor and Course Coordinator	70% of student achieve 70% of the CLO	- Critical and analytical thinking - Written communication	
Present and demonstrate the design and prototype of the project.	- Case studies - Group work - Team based learning	- Examiner - Examining Committee	70% of student achieve 70% of the CLO	- Critical and analytical thinking	
Work as an efficient member of a team on the design and development of the project.	- Group work - Problem-solving - Team based learning - Teamwork	- Examiner - Examining Committee - Supervisor and Course Coordinator	70% of student achieve 70% of the CLO	- Critical and analytical thinking - Teamwork or collaboration - Leadership skills	

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
				- Intellectual curiosity and lifelong learning	

Scope and Nature of Assessment Tools:

Examiner: they will assess different components, such as: Project Plan, Project Design & Solution Quality, Integration of IT Technologies, User Testing & Evaluation, Oral Presentation + Demo, Report)

Supervisor and Course Coordinator: will assess the final project deliverables

Examining Committee: they will assess different components, such as: Project Plan, Project Design & Solution Quality, Integration of IT Technologies, User Testing & Evaluation, Oral Presentation + Demo)

Teaching and Learning Resources:

Textbook(s):

Thonney, T.. (2015). Academic Writing: Concepts and Connections. 1st ed. Oxford University Press.

Silvia, P. J.. (2018). How to Write a Lot: A Practical Guide to Productive Academic Writing. 2nd ed. APA LifeTools.

Reinhart, S. M.. (2013). Giving Academic Presentations. 2nd ed. University of Michigan Press.

Schwabish, J.. (2016). Better Presentations: A Guide for Scholars, Researchers, and Wonks. 1st ed. Columbia University Press.

Other Resources:

Required Teaching Resources:

- University software and lab equipment.
- Other software and equipment required by the projects.

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Notes:

Teaching and Learning:

The course consists of few lectures at the beginning, then the students work on a group project. Students undergo the following phases: literature review, searching, identifying a problem, design, analysis, report writing, and prototyping.

In-class Activities:

Students work as a team on a Multimedia project. The size of the team depends on the complexity of the project. Work of a project is divided among member students. Students are expected to collaborate and communicate to integrate their works. At the end of the course, students submit a report and demo the prototype of the multimedia system. The project and the report are evaluated based on its depth as shown by the content of the report, organization, conclusion, demonstration and presentation.

Grading rubrics:

Examining Committee - Junior Project Evaluation Form

Trait	Unacceptable	Below Expectation	Meets Expectation with Improvements	Meets Expectation	Weight	Student Mark (out of 5)
	[0-1.25)	[1.25-3.25)	[3.25-4.5)	[4.5-5]		
Project Plan	Fails to identify technical or logistical challenges. Schedule and milestones unclear or unrealistic.	Identifies some challenges without adequate mitigation. Timeline loosely aligned.	Identifies most challenges. Timeline and milestones align reasonably but could be refined.	Comprehensive planning with clear challenges, milestones, schedule, and risk mitigation. Professional approach.	25 %	
Project Design & Solution Quality	Solution/design does not meet objectives. No structured approach or requirement alignment.	Solution/design partially meets objectives. Weak justification or lack of consideration for alternatives.	Meets objectives. Sound design process with some improvements needed in alignment or clarity.	Clearly meets/exceeds objectives with a well-justified, systematic design approach linked to requirements.	25 %	
Integration of IT Technologies	No appropriate technology used. Lacks understanding and justification of tools.	Basic or incorrect use of tools. Weak alignment between technology and solution needs.	Appropriate use of tools and technologies with some justification. Opportunity for optimization.	Professional use of modern tools with clear justification, such as (if applicable), version control (GitHub, GitLab, etc.), cloud/DevOps, frontend and backend development technologies, databases, and AI.	20 %	

User Testing & Evaluation Plan	No testing or feedback. No meaningful metrics used to evaluate performance or usability.	Limited testing with undefined or poorly explained metrics.	Adequate testing and evaluation using some relevant metrics	Complete testing and evaluation using appropriate metrics.	5%	
Oral Presentation + Demo	Unclear delivery. Poor timing, unprepared, or unable to handle questions.	Relies heavily on notes with unclear explanations. Weak question response. Weak demo.	Clear and mostly confident delivery with minor improvements needed in flow or interaction.	Confident, well-structured, engaging delivery. Responds to questions accurately. Uses time effectively.	25%	
					Total	/5

Examiner

Trait	Unacceptable	Below Expectation	Meets Expectation with Improvements	Meets Expectation	Weight	Student Mark (out of 5)
	[0-1.25]	[1.25-3.25]	[3.25-4.5]	[4.5-5]		
Project Plan	Fails to identify technical or logistical challenges. Schedule and milestones unclear or unrealistic.	Identifies some challenges without adequate mitigation. Timeline loosely aligned.	Identifies most challenges. Timeline and milestones align reasonably but could be refined.	Comprehensive planning with clear challenges, milestones, schedule, and risk mitigation. Professional approach.	20%	
Project Design & Solution Quality	Solution does not meet objectives. No structured approach or requirement alignment.	Design partially meets objectives. Weak justification or lack of consideration for alternatives.	Meets objectives. Sound design process with some improvements needed in alignment or clarity.	Clearly meets/exceeds objectives with a well-justified, systematic design approach linked to requirements.	15%	

Integration of IT Technologies	No appropriate technology used. Lacks understanding and justification of tools.	Basic or incorrect use of tools. Weak alignment between technology and solution needs.	Appropriate use of tools and technologies with some justification. Opportunity for optimization.	Professional use of modern tools with clear justification, such as (if applicable), version control (GitHub, GitLab, etc.), cloud/DevOps, frontend and backend development technologies, databases, and AI.	15 %	4.5
User Testing & Evaluation	No testing or feedback. No meaningful metrics used to evaluate performance or usability.	Limited testing with undefined or poorly explained metrics.	Adequate testing and evaluation using some relevant metrics	Complete testing and evaluation using appropriate metrics.	5%	5
Oral Presentation + Demo	Unclear delivery. Poor timing, unprepared, or unable to handle questions.	Relies heavily on notes with unclear explanations. Weak question response. Weak demo.	Clear and mostly confident delivery with minor improvements needed in flow or interaction.	Confident, well-structured, engaging delivery. Responds to questions accurately. Uses time effectively.	20 %	4
Final Report	Report lacks organization. Hard to follow logic. Diagrams missing or inconsistent. Grammar and spelling errors present. Possible plagiarism.	Report generally clear, with minor inconsistencies in logic or style. Occasional grammar issues but understandable. Visuals generally support content.	Well-organized, clearly written report. Minor refinements needed in logic flow, clarity, or diagram use. Good structure and support.	Highly polished and organized report. Clear logical flow, precise language, accurate diagrams. Free of grammatical and spelling errors. Proper citation.	25 %	
					Total	/5